DIVIDE AND CONQUER

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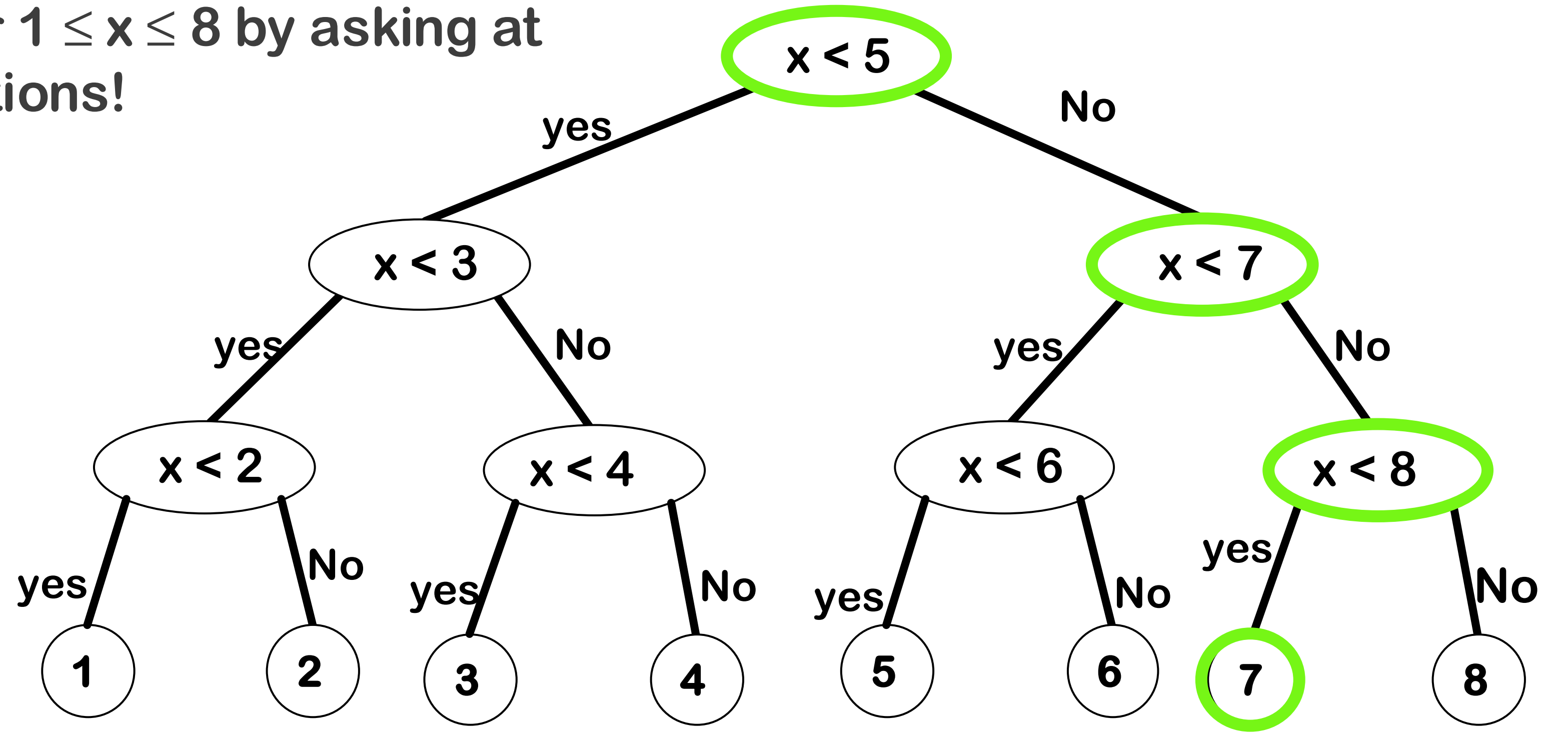
NJIT

GUESS A NUMBER!

- Guess an integer $1 \leq x \leq 8$ by asking at most three questions!



GUESS A NUMBER!

- Guess an integer $1 \leq x \leq 16$ by asking at most three questions!
- Guess 7

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	x<9??
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	x<5??
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	x<7??
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	x<8??
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	

INTRODUCTION

Napoleon

DEFINITION

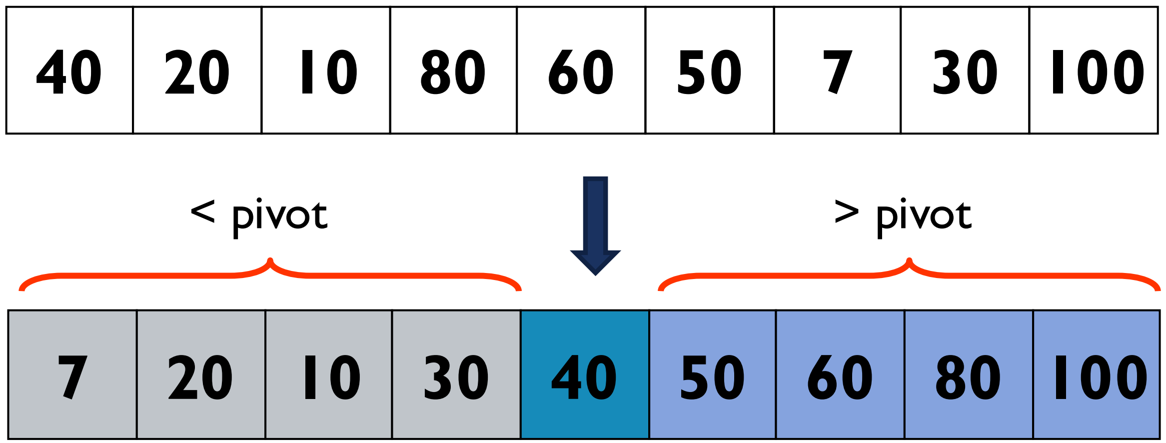
- **Divide** the problem into a number of subproblems that are smaller instances of the same problem.
- **Conquer** the subproblems by solving them recursively. If they are small enough, solve the subproblems as base cases.
- **Combine** the solutions to the subproblems into the solution for the original problem.

EXAMPLE 1: QUICKSORT

- **Divide** the list into two smaller list: less than pivot and greater than pivot.
- **Conquer** the subproblems by sorting the sublists recursively.
- **Combine** the sorted sublists to obtain the sorted main list.

QUICKSORT: DIVIDE

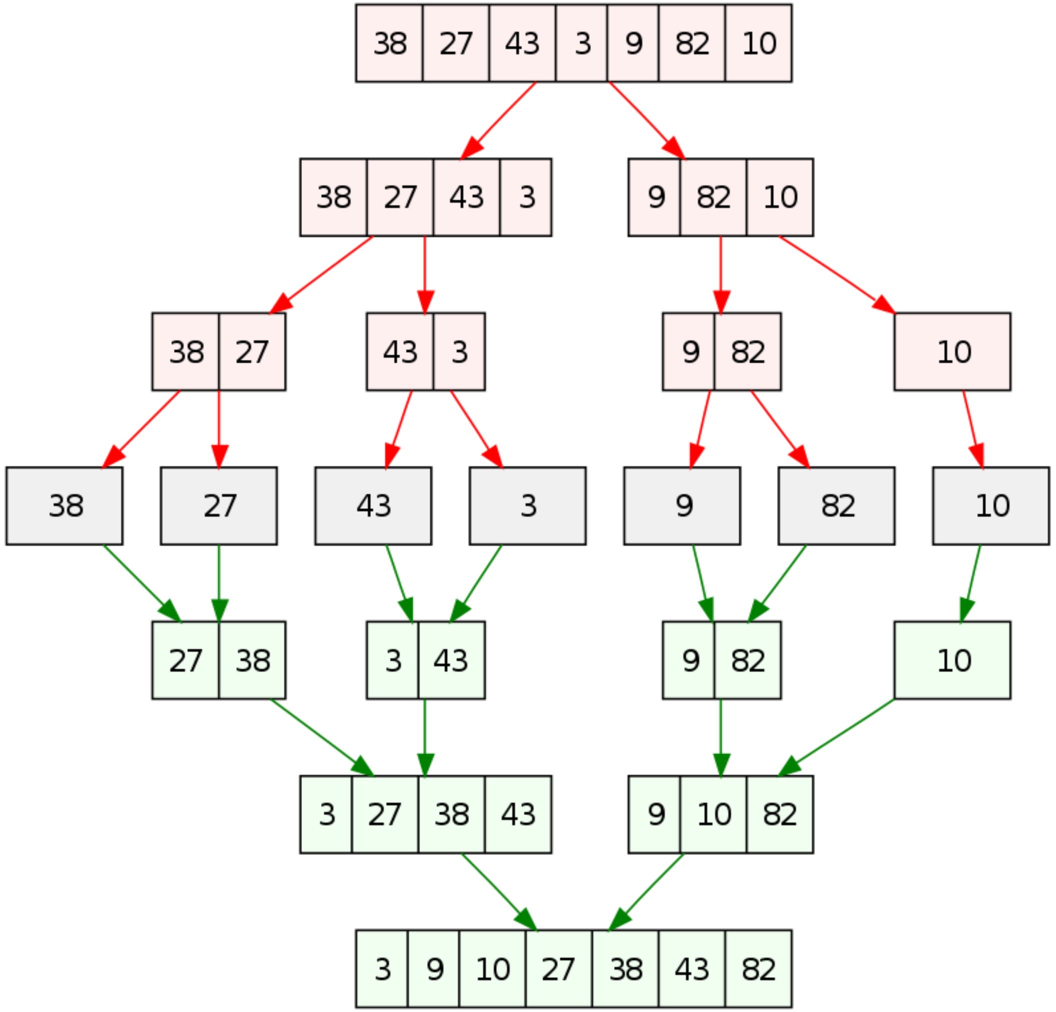
- **Divide** the list into two smaller list: less than pivot and greater than pivot.



EXAMPLE 2: MERGESORT

- **Divide** the list into two equal size sublists.
- **Conquer** the subproblems by sorting the lists recursively.
- **Combine(Merge)** the sorted lists to obtain the sorted list.

MERGESORT



38 27 43 3 9 82 10
38 27 43 3
9 82 10
38 27
43 3
9 82
10
38
27
43
3
9
82
10
27 38
3 43
9 82
10
3 27 38 43
9 10 82
3 9 10 27 38 43 82

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

$$\begin{bmatrix} 7 & 1 & 4 & 5 & 6 & 7 & 8 & 9 \\ 1 & 2 & 4 & 6 & 3 & 5 & 8 & 3 \\ 5 & 3 & 6 & 7 & 4 & 1 & 3 & 2 \\ 0 & 4 & 0 & 6 & 7 & 5 & 7 & 2 \\ 0 & 4 & 6 & 7 & 0 & 1 & 6 & 3 \\ 8 & 9 & 4 & 6 & 3 & 5 & 8 & 4 \\ 4 & 6 & 6 & 7 & 4 & 1 & 3 & 6 \\ 2 & 0 & 0 & 6 & 7 & 5 & 7 & 0 \end{bmatrix}_{nxn}$$

n^2

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

$$\begin{bmatrix} 7 & 5 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 4 \\ 8 & 6 \end{bmatrix} = \begin{bmatrix} 7+1 & 5+4 \\ 3+8 & 2+6 \end{bmatrix}$$

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

$$
\begin{matrix}
7 & 1 & 4 & 5 & 6 & 7 & 8 & 9 \\
1 & 2 & 4 & 6 & 3 & 5 & 8 & 3 \\
5 & 3 & 6 & 7 & 4 & 1 & 3 & 2 \\
0 & 4 & 0 & 6 & 7 & 5 & 7 & 2 \\
0 & 4 & 6 & 7 & 0 & 1 & 6 & 3 \\
8 & 9 & 4 & 6 & 3 & 5 & 8 & 4 \\
4 & 6 & 6 & 7 & 4 & 1 & 3 & 6 \\
2 & 0 & 0 & 6 & 7 & 5 & 7 & 0
\end{matrix}
\quad + \quad
\begin{matrix}
4 & 1 & 3 & 2 & 4 & 0 & 6 & 7 \\
7 & 5 & 7 & 2 & 7 & 6 & 7 & 0 \\
0 & 1 & 6 & 3 & 0 & 4 & 6 & 3 \\
3 & 5 & 8 & 4 & 3 & 6 & 7 & 4 \\
4 & 1 & 3 & 6 & 4 & 0 & 6 & 7 \\
7 & 5 & 7 & 0 & 7 & 6 & 7 & 0 \\
0 & 6 & 7 & 0 & 4 & 0 & 6 & 7 \\
6 & 7 & 0 & 6 & 7 & 6 & 7 & 0
\end{matrix}
\quad = \quad
\begin{matrix}
11 & 2 & 7 & 7 & 10 & 7 & 14 & 16 \\
8 & 7 & 11 & 8 & 10 & 11 & 15 & 6 \\
5 & 4 & 12 & 10 & 4 & 5 & 9 & 5 \\
3 & 9 & 8 & 10 & 10 & 11 & 14 & 6 \\
4 & 5 & 9 & 13 & 4 & 1 & 12 & 10 \\
15 & 14 & 11 & 6 & 10 & 11 & 15 & 4 \\
4 & 12 & 13 & 7 & 8 & 1 & 9 & 13 \\
8 & 7 & 0 & 12 & 14 & 11 & 14 & 0
\end{matrix}
$$

n^2

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

$$\begin{bmatrix} \boxed{7 \quad 5} \\ 3 \quad 2 \end{bmatrix} \times \begin{bmatrix} \boxed{\begin{matrix} 1 \\ 8 \end{matrix}} & \begin{matrix} 4 \\ 6 \end{matrix} \end{bmatrix} = \begin{bmatrix} \boxed{7\text{x}1+5\text{x}8} & 7\text{x}4+5\text{x}6 \\ 3\text{x}1+2\text{x}8 & 3\text{x}4+2\text{x}6 \end{bmatrix}$$

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

$$\begin{bmatrix} \boxed{7 \quad 5} \\ 3 \quad 2 \end{bmatrix} \times \begin{bmatrix} 1 & \boxed{4} \\ 8 & \boxed{6} \end{bmatrix} = \begin{bmatrix} 7\text{x}1+5\text{x}8 & \boxed{7\text{x}4+5\text{x}6} \\ 3\text{x}1+2\text{x}8 & 3\text{x}4+2\text{x}6 \end{bmatrix}$$

EXAMPLE 3: STRASSEN’S MATRIX MULTIPLICATION

$$
\begin{matrix}
7 & 1 & 4 & 5 & 6 & 7 & 8 & 9 \\
1 & 2 & 4 & 6 & 3 & 5 & 8 & 3 \\
5 & 3 & 6 & 7 & 4 & 1 & 3 & 2 \\
0 & 4 & 0 & 6 & 7 & 5 & 7 & 2 \\
0 & 4 & 6 & 7 & 0 & 1 & 6 & 3 \\
8 & 9 & 4 & 6 & 3 & 5 & 8 & 4 \\
4 & 6 & 6 & 7 & 4 & 1 & 3 & 6 \\
2 & 0 & 0 & 6 & 7 & 5 & 7 & 0
\end{matrix}
\times
\begin{matrix}
4 & 1 & 3 & 2 & 4 & 0 & 6 & 7 \\
7 & 5 & 7 & 2 & 7 & 6 & 7 & 0 \\
0 & 1 & 6 & 3 & 0 & 4 & 6 & 3 \\
3 & 5 & 8 & 4 & 3 & 6 & 7 & 4 \\
4 & 1 & 3 & 6 & 4 & 0 & 6 & 7 \\
7 & 5 & 7 & 0 & 7 & 6 & 7 & 0 \\
0 & 6 & 7 & 0 & 4 & 0 & 6 & 7 \\
6 & 7 & 0 & 6 & 7 & 6 & 7 & 0
\end{matrix}
=
\begin{matrix}
177 & 193 & 215 & 138 & 218 & 148 & 304 & 179 \\
101 & 142 & 189 & 78 & 136 & 112 & 208 & 120 \\
97 & 102 & 168 & 98 & 111 & 102 & 199 & 130 \\
121 & 138 & 181 & 86 & 151 & 102 & 203 & 122 \\
74 & 123 & 169 & 72 & 101 & 114 & 177 & 88 \\
184 & 191 & 259 & 112 & 220 & 160 & 306 & 169 \\
138 & 144 & 186 & 126 & 156 & 144 & 242 & 123 \\
89 & 106 & 159 & 70 & 117 & 66 & 173 & 136
\end{matrix}
$$

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

7	1	4	5	6	7	8	9
1	2	4	6	3	5	8	3
5	3	6	7	4	1	3	2
0	4	0	6	7	5	7	2
0	4	6	7	0	1	6	3
8	9	4	6	3	5	8	4
4	6	6	7	4	1	3	6
2	0	0	6	7	5	7	0

x

4	1	3	2	4	0	6	7
7	5	7	2	7	6	7	0
0	1	6	3	0	4	6	3
3	5	8	4	3	6	7	4
4	1	3	6	4	0	6	7
7	5	7	0	7	6	7	0
0	6	7	0	4	0	6	7
6	7	0	6	7	6	7	0

=

177	193	215	138	218	148	304	179
101	142	189	78	136	112	208	120
97	102	168	98	111	102	199	130
121	138	181	86	151	102	203	122
74	123	169	72	101	114	177	88
184	191	259	112	220	160	306	169
138	144	186	126	156	144	242	123
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7	1	4	5	6	7	8	9
1	2	4	6	3	5	8	3
5	3	6	7	4	1	3	2
0	4	0	6	7	5	7	2
0	4	6	7	0	1	6	3
8	9	4	6	3	5	8	4
4	6	6	7	4	1	3	6
2	0	0	6	7	5	7	0

x

4	1	3	2	4	0	6	7
7	5	7	2	7	6	7	0
0	1	6	3	0	4	6	3
3	5	8	4	3	6	7	4
4	1	3	6	4	0	6	7
7	5	7	0	7	6	7	0
0	6	7	0	4	0	6	7
6	7	0	6	7	6	7	0

=

177	193	215	138	218	148	304	179
101	142	189	78	136	112	208	120
97	102	168	98	111	102	199	130
121	138	181	86	151	102	203	122
74	123	169	72	101	114	177	88
184	191	259	112	220	160	306	169
138	144	186	126	156	144	242	123
89	106	159	70	117	66	173	136

$$n-1 \rightarrow n^2(n-1) = O(n^3)$$

EXAMPLE 3: MATRIX MULTIPLICATION

$$C_{nxn} = A_{nxn} \times B_{nxn}$$

```
for(i=0;i<n;i++)
     for(j=0;j<n;j++)
    {
     C[i][j]=0;
         for(k=0;k<n;k++)
          C[i][j]+=A[i][k]*B[k][j];

    }
```

$$O(n^3) \rightarrow O(n^?)$$

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

$$
\begin{array}{cccc|cccc}
7 & 1 & 4 & 5 & 6 & 7 & 8 & 9 \\
1 & 2 & 4 & 6 & 3 & 5 & 8 & 3 \\
5 & 3 & 6 & 7 & 4 & 1 & 3 & 2 \\
0 & 4 & 0 & 6 & 7 & 5 & 7 & 2 \\
\hline
0 & 4 & 6 & 7 & 0 & 1 & 6 & 3 \\
8 & 9 & 4 & 6 & 3 & 5 & 8 & 4 \\
4 & 6 & 6 & 7 & 4 & 1 & 3 & 6 \\
2 & 0 & 0 & 6 & 7 & 5 & 7 & 0
\end{array}
\times
\begin{array}{cccc|cccc}
4 & 1 & 3 & 2 & 4 & 0 & 6 & 7 \\
7 & 5 & 7 & 2 & 7 & 6 & 7 & 0 \\
0 & 1 & 6 & 3 & 0 & 4 & 6 & 3 \\
3 & 5 & 8 & 4 & 3 & 6 & 7 & 4 \\
\hline
4 & 1 & 3 & 6 & 4 & 0 & 6 & 7 \\
7 & 5 & 7 & 0 & 7 & 6 & 7 & 0 \\
0 & 6 & 7 & 0 & 4 & 0 & 6 & 7 \\
6 & 7 & 0 & 6 & 7 & 6 & 7 & 0
\end{array}
= ?
$$

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

7	1	4	5	6	7	8	9
1	2	4	6	3	5	8	3
5	3	6	7	4	1	3	2
0	4	0	6	7	5	7	2
0	4	6	7	0	1	6	3
8	9	4	6	3	5	8	4
4	6	6	7	4	1	3	6
2	0	0	6	7	5	7	0

x

4	1	3	2	4	0	6	7
7	5	7	2	7	6	7	0
0	1	6	3	0	4	6	3
3	5	8	4	3	6	7	4
4	1	3	6	4	0	6	7
7	5	7	0	7	6	7	0
0	6	7	0	4	0	6	7
6	7	0	6	7	6	7	0

=

177	193	215	138	218	148	304	179
101	142	189	78	136	112	208	120
97	102	168	98	111	102	199	130
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$$
\begin{matrix}
7 & 1 & 4 & 5 & 6 & 7 & 8 & 9 \\
1 & 2 & 4 & 6 & 3 & 5 & 8 & 3 \\
5 & 3 & 6 & 7 & 4 & 1 & 3 & 2 \\
0 & 4 & 0 & 6 & 7 & 5 & 7 & 2 \\
0 & 4 & 6 & 7 & 0 & 1 & 6 & 3 \\
8 & 9 & 4 & 6 & 3 & 5 & 8 & 4 \\
4 & 6 & 6 & 7 & 4 & 1 & 3 & 6 \\
2 & 0 & 0 & 6 & 7 & 5 & 7 & 0
\end{matrix}
\times
\begin{matrix}
4 & 1 & 3 & 2 & 4 & 0 & 6 & 7 \\
7 & 5 & 7 & 2 & 7 & 6 & 7 & 0 \\
0 & 1 & 6 & 3 & 0 & 4 & 6 & 3 \\
3 & 5 & 8 & 4 & 3 & 6 & 7 & 4 \\
4 & 1 & 3 & 6 & 4 & 0 & 6 & 7 \\
7 & 5 & 7 & 0 & 7 & 6 & 7 & 0 \\
0 & 6 & 7 & 0 & 4 & 0 & 6 & 7 \\
6 & 7 & 0 & 6 & 7 & 6 & 7 & 0
\end{matrix}
=
\begin{matrix}
177 & 193 & 215 & 138 & 218 & 148 & 304 & 179 \\
101 & 142 & 189 & 78 & 136 & 112 & 208 & 120 \\
97 & 102 & 168 & 98 & 111 & 102 & 199 & 130 \\
121 & 138 & 181 & 86 & 151 & 102 & 203 & 122 \\
74 & 123 & 169 & 72 & 101 & 114 & 177 & 88 \\
184 & 191 & 259 & 112 & 220 & 160 & 306 & 169 \\
138 & 144 & 186 & 126 & 156 & 144 & 242 & 123 \\
89 & 106 & 159 & 70 & 117 & 66 & 173 & 136
\end{matrix}
$$

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

7	1	4	5	6	7	8	9
1	2	4	6	3	5	8	3
5	3	6	7	4	1	3	2
0	4	0	6	7	5	7	2
0	4	6	7	0	1	6	3
8	9	4	6	3	5	8	4
4	6	6	7	4	1	3	6
2	0	0	6	7	5	7	0

x

4	1	3	2	4	0	6	7
7	5	7	2	7	6	7	0
0	1	6	3	0	4	6	3
3	5	8	4	3	6	7	4
4	1	3	6	4	0	6	7
7	5	7	0	7	6	7	0
0	6	7	0	4	0	6	7
6	7	0	6	7	6	7	0

=

177	193	215	138	218	148	304	179
101	142	189	78	136	112	208	120
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121	138	181	86	151	102	203	122
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7	1	4	5	6	7	8	9
1	2	4	6	3	5	8	3
5	3	6	7	4	1	3	2
0	4	0	6	7	5	7	2
0	4	6	7	0	1	6	3
8	9	4	6	3	5	8	4
4	6	6	7	4	1	3	6
2	0	0	6	7	5	7	0

x

4	1	3	2	4	0	6	7
7	5	7	2	7	6	7	0
0	1	6	3	0	4	6	3
3	5	8	4	3	6	7	4
4	1	3	6	4	0	6	7
7	5	7	0	7	6	7	0
0	6	7	0	4	0	6	7
6	7	0	6	7	6	7	0

=

177	193	215	138	218	148	304	179
101	142	189	78	136	112	208	120
97	102	168	98	111	102	199	130
121	138	181	86	151	102	203	122
74	123	169	72	101	114	177	88
184	191	259	112	220	160	306	169
138	144	186	126	156	144	242	123
89	106	159	70	117	66	173	136

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \times \begin{pmatrix} E & F \\ G & H \end{pmatrix} = \begin{pmatrix} AE+BG & AF+BH \\ CE+DG & CF+DH \end{pmatrix}$$

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

$$\begin{pmatrix} A & \cdots & +BG & \boxed{AF+BH} \\ C & \cdots & +DG & CF+DH \end{pmatrix}$$

$P_1 + P_2 =$
$A(F-H) + (A+B)H =$
$AF - AH + AH + BH =$
$AF + BH$

Trick:

$P_1 = A \cdot (F-H)$
$P_2 = (A+B) \cdot H$
$P_3 = (C+D) \cdot E$
$P_4 = D \cdot (G-E)$
$P_5 = (A+D) \cdot (E+$
$P_6 = (B-D) \cdot (G+H)$
$P_7 = (A-C) \cdot (E+F)$

$AE+BG = P_5 + P_4 - P_2 + P_6$
$AF+BH = P_1 + P_2$
$CE+DG = P_3 + P_4$
$CF+DH = P_5 + P_1 - P_3 - P_7$

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \times \begin{pmatrix} E & F \\ G & H \end{pmatrix} = \begin{pmatrix} AE+BG & AF+BH \\ CE+DG & CF+DH \end{pmatrix}$$

Trick:

$P_1 = A \cdot (F-H)$

$P_2 = (A+B) \cdot H$

$P_3 = (C+D) \cdot E$

$P_4 = D \cdot (G-E)$

$P_5 = (A+D) \cdot (E+H)$

$P_6 = (B-D) \cdot (G+H)$

$P_7 = (A-C) \cdot (E+F)$

$AE+BG = P_5 + P_4 - P_2 + P_6$

$AF+BH = P_1 + P_2$

$CE+DG = P_3 + P_4$

$CF+DH = P_5 + P_1 - P_3 - P_7$

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \times \begin{pmatrix} E & F \\ G & H \end{pmatrix} = \begin{bmatrix} P_5 + P_4 - P_2 + P_6 & P_1 + P_2 \\ P_3 + P_4 & P_5 + P_1 - P_3 - P_7 \end{bmatrix}$$

Trick:

$P_1 = A \cdot (F-H)$

$P_2 = (A+B) \cdot H$

$P_3 = (C+D) \cdot E$

$P_4 = D \cdot (G-E)$

$P_5 = (A+D) \cdot (E+H)$

$P_6 = (B-D) \cdot (G+H)$

$P_7 = (A-C) \cdot (E+F)$

$AE+BG = P_5 + P_4 - P_2 + P_6$

$AF+BH = P_1 + P_2$

$CE+DG = P_3 + P_4$

$CF+DH = P_5 + P_1 - P_3 - P_7$

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

$$
\left[\begin{array}{cccc|cccc}
7 & 1 & 4 & 5 & 6 & 7 & 8 & 9 \\
1 & 2 & 4 & 6 & 3 & 5 & 8 & 3 \\
5 & 3 & 6 & 7 & 4 & 1 & 3 & 2 \\
0 & 4 & 0 & 6 & 7 & 5 & 7 & 2 \\
\hline
0 & 4 & 6 & 7 & 0 & 1 & 6 & 3 \\
8 & 9 & 4 & 6 & 3 & 5 & 8 & 4 \\
4 & 6 & 6 & 7 & 4 & 1 & 3 & 6 \\
2 & 0 & 0 & 6 & 7 & 5 & 7 & 0
\end{array}\right]
\times
\left[\begin{array}{cccc|cccc}
4 & 1 & 3 & 2 & 4 & 0 & 6 & 7 \\
7 & 5 & 7 & 2 & 7 & 6 & 7 & 0 \\
0 & 1 & 6 & 3 & 0 & 4 & 6 & 3 \\
3 & 5 & 8 & 4 & 3 & 6 & 7 & 4 \\
\hline
4 & 1 & 3 & 6 & 4 & 0 & 6 & 7 \\
7 & 5 & 7 & 0 & 7 & 6 & 7 & 0 \\
0 & 6 & 7 & 0 & 4 & 0 & 6 & 7 \\
6 & 7 & 0 & 6 & 7 & 6 & 7 & 0
\end{array}\right]
=?
$$

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

```
Matrix Strassen(A, B, n):
       if (n==1) return A*B

       a, b, c, d = split(A)
       e, f, g, h = split(B)

       p1 = Strassen(a, f - h, n/2)
       p2 = Strassen(a + b, h, n/2)
       p3 = Strassen(c + d, e, n/2)
       p4 = Strassen(d, g - e, n/2)
       p5 = Strassen(a + d, e + h, n/2)
       p6 = Strassen(b - d, g + h, n/2)
       p7 = Strassen(a - c, e + f, n/2)

       c11 = p5 + p4 - p2 + p6;
       c12 = p1 + p2
       c21 = p3 + p4
       c22 = p1 + p5 - p3 - p7
       return C
```

```
Matrix Strassen(A, B, n)
    if (n==1) return A*B
    a, b, c, d = split(A)
    e, f, g, h = split(B)
    p1 = Strassen(a, f - h, n/2)
    p2 = Strassen(a + b, h, n/2)
    p3 = Strassen(c + d, e, n/2)
    p4 = Strassen(d, g - e, n/2)
    p5 = Strassen(a + d, e + h, n/2)
    p6 = Strassen(b - d, g + h, n/2)
    p7 = Strassen(a - c, e + f, n/2)
    c11 = p5 + p4 - p2 + p6
    c12 = p1 + p2
    c21 = p3 + p4
    c22 = p1 + p5 - p3 - p7
    return C
```

$\rightarrow T(n)$

$\rightarrow T(n) = 1$

$\rightarrow O(n^2)$

$\rightarrow T\left(\frac{n}{2}\right) + \left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$

$\rightarrow T\left(\frac{n}{2}\right) + \left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$

$\rightarrow T\left(\frac{n}{2}\right) + \left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$

$\rightarrow T\left(\frac{n}{2}\right) + \left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$

$\rightarrow T\left(\frac{n}{2}\right) + 2\left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$

$\rightarrow T\left(\frac{n}{2}\right) + 2\left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$

$\rightarrow T\left(\frac{n}{2}\right) + 2\left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$

$\rightarrow 3\left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$

$\rightarrow \left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$

$\rightarrow \left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$

$\rightarrow 3\left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$

$$\boxed{T(n) = O(n^2) + 7T\left(\frac{n}{2}\right) + 18\left(\frac{n}{2}\right)\left(\frac{n}{2}\right)}$$

STRASSEN: TIME COMPLEXITY-MASTER THEOREM

$$T(n) = O(n^2) + 7T\left(\frac{n}{2}\right)$$

$$a = 7, b = 2, k = 2, p = 0$$

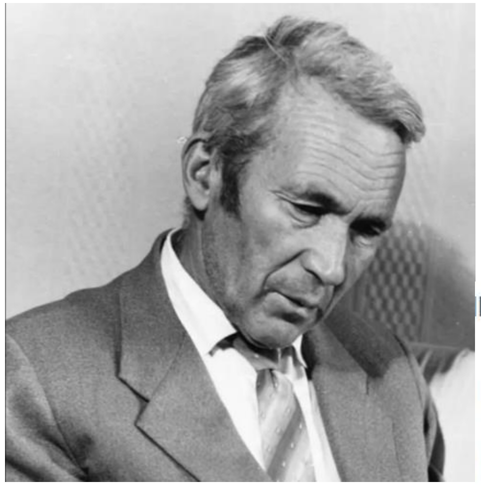
$$T(n) = O\left(n^{\log_2 7}\right) = O(n^{2.81})$$

$$T(n) = a\,T\left(\frac{n}{b}\right) + \theta(n^k \log^p n)$$

$$if\ a > b^k, then\ T(n) = \theta\left(n^{\log_b a}\right)$$

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$
\begin{array}{ccccc}
 & & & 2 & 5 \\
 & & \times & 6 & 3 \\
\hline
 & & & 1 & 5 \\
 & & & 6 & 0 \\
+ & & 3 & 0 & 0 \\
 & 1 & 2 & 0 & 0 \\
\hline
 & 1 & 5 & 7 & 5
\end{array}
$$



Andrey Kolmogorov

ls in [2] what happened:

In the autumn of 1960 a seminar on problems in cybernetics was held at the Faculty of Mechanics and Mathematics of Moscow University under the direction of Kolmogorov. There Kolmogorov stated the n^2 conjecture and proposed some problems concerning the complexity of the solution of linear systems of equations and some similar calculations. I began to think actively about the n^2 conjecture, and in less than a week I found that the algorithm with which I was planning to derive a lower bound for $M(n)$ gave a bound of the form

$$M(n) = O(n^{\log_2 3}), \qquad \log_2 3 = 1.5849\ldots$$

Anatoly A. Karatsuba

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

- Storing an n-digit large integer: $\mathbf{8,256,734}$

8	2	5	6	7	3	4

- Multiply be a power of 10: $\mathbf{8,256,734} \times \mathbf{10^4}$

8	2	5	6	7	3	4	0	0	0	0

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

- Adding two large integers: $\mathbf{850368000000} + \mathbf{7519802000} = ?$

8	5	0	3	6	8	0	0	0	0	0	0
		7	5	1	9	8	0	2	0	0	0

carry

0

L.I.M: SUMMATION

8	5	0	3	6	8	0	0	0	0	0	0

7	5	1	9	8	0	2	0	0	0

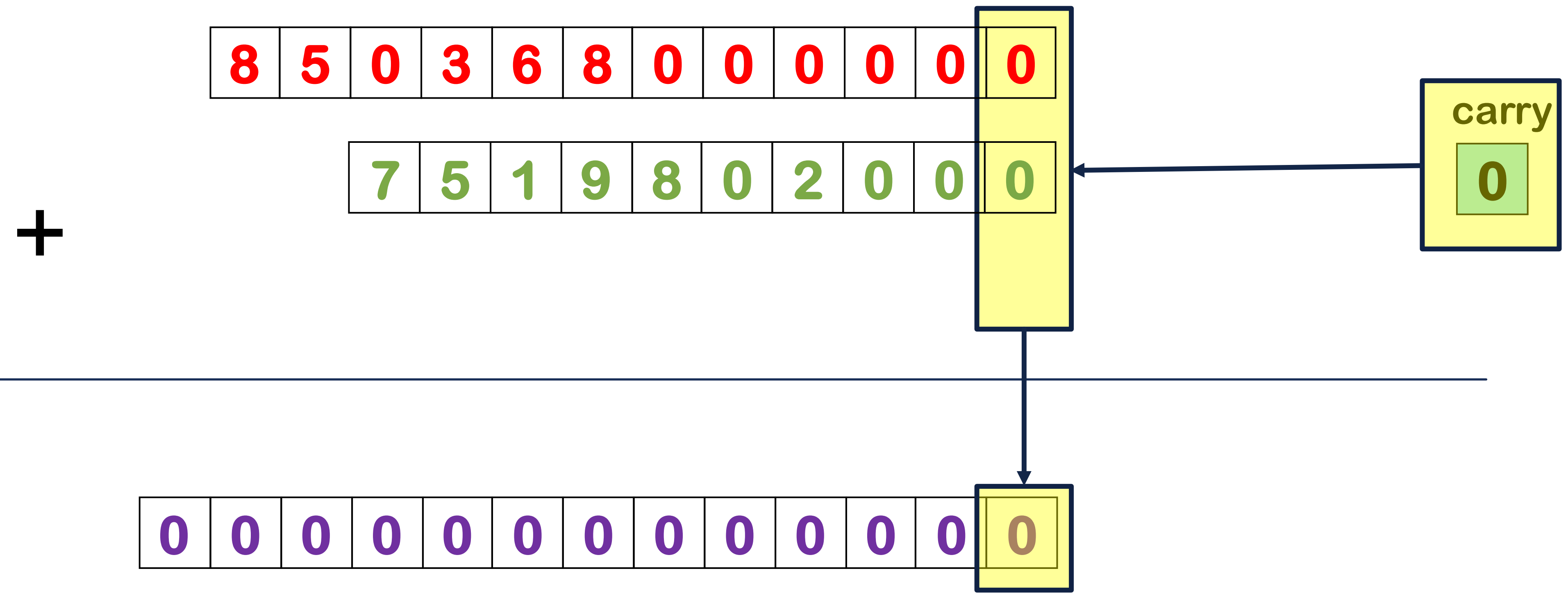
+

carry

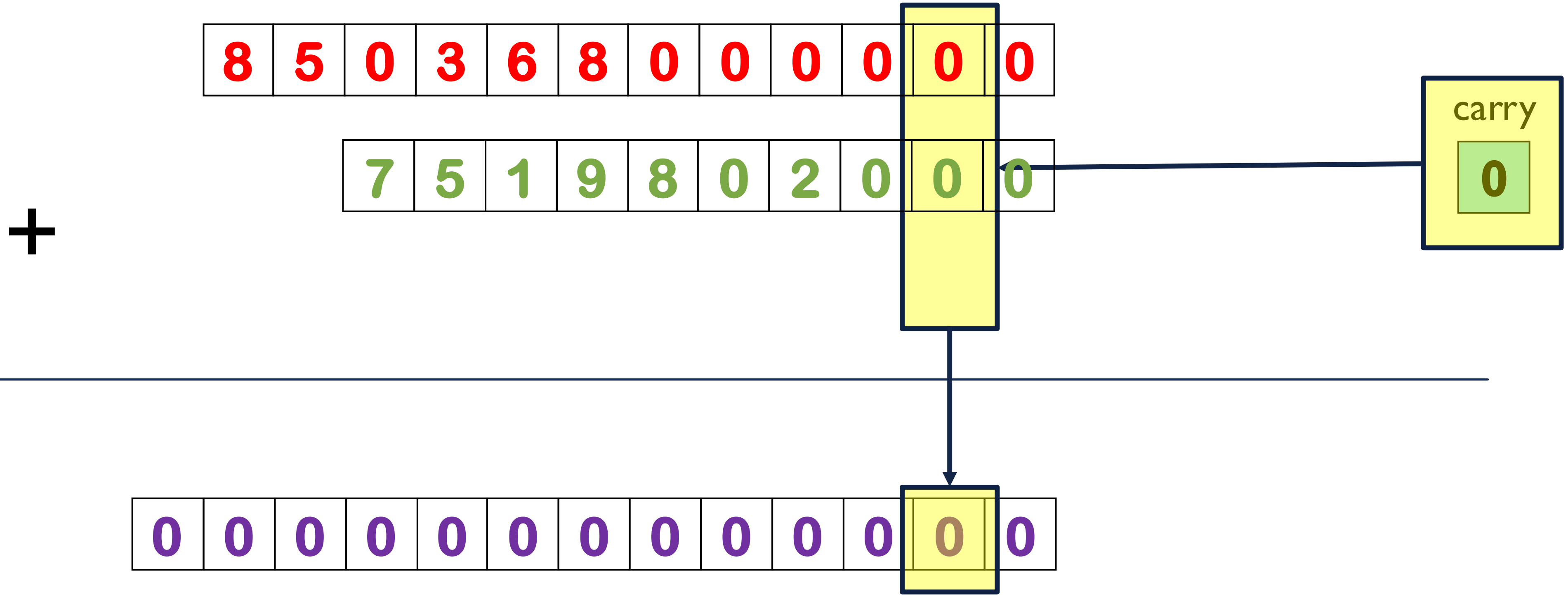
0

0	0	0	0	0	0	0	0	0	0	0	0	0

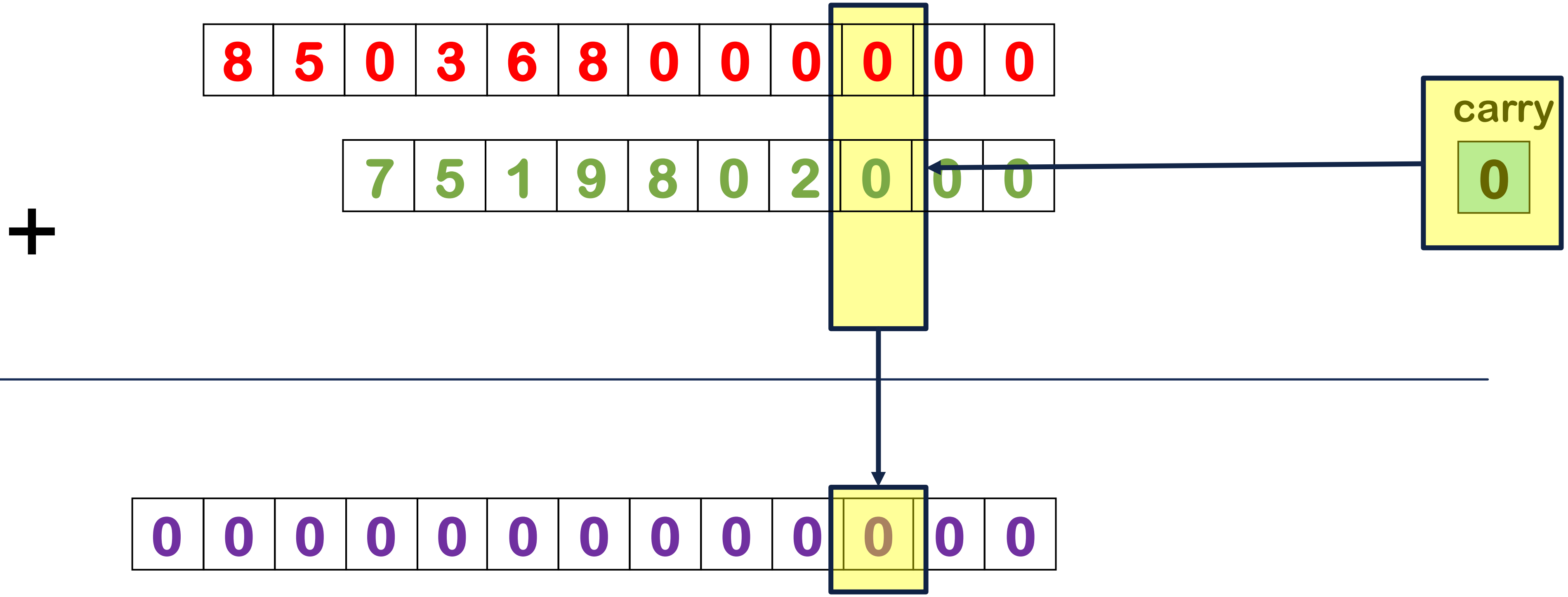
L.I.M: SUMMATION



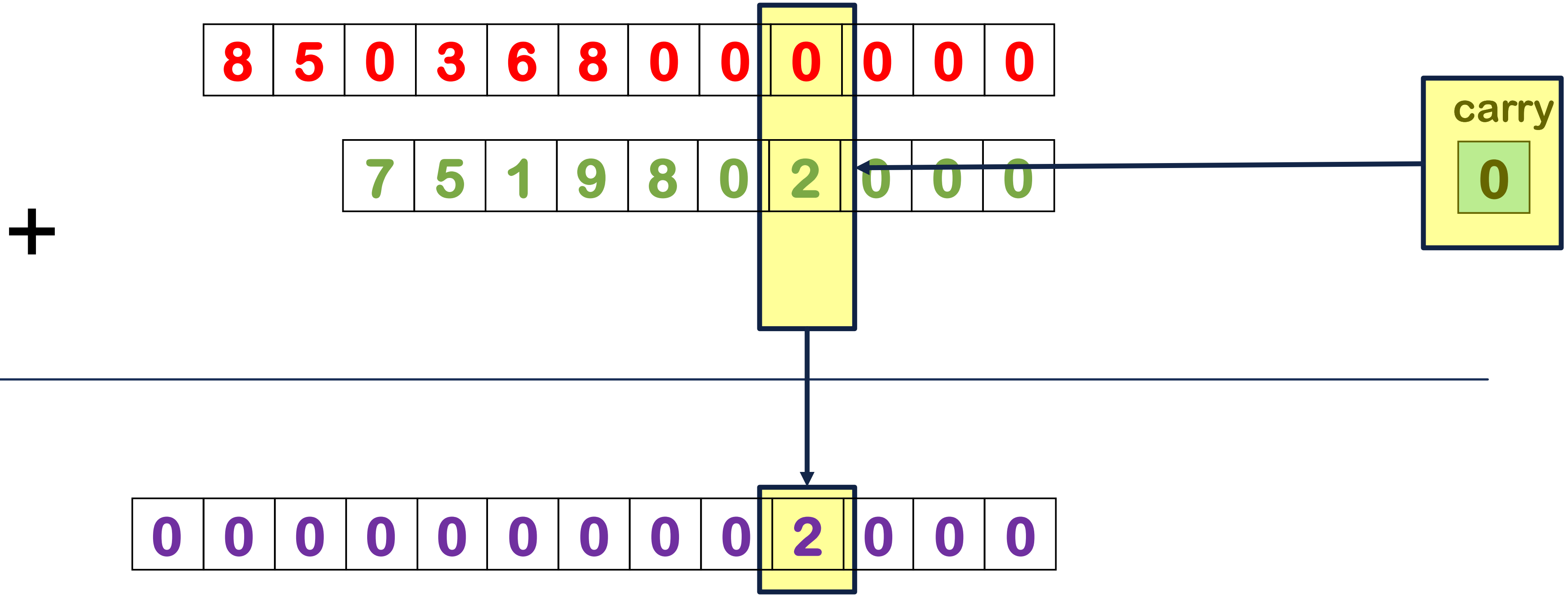
L.I.M: SUMMATION



L.I.M: SUMMATION



L.I.M: SUMMATION



	8	5	0	3	6	8	0	0	0	0	0	0
			7	5	1	9	8	0	2	0	0	0
+												
0	0	0	0	0	8	7	8	0	2	0	0	0

carry

1

L.I.M: SUMMATION

8	5	0	3	6	8	0	0	0	0	0	0

\+

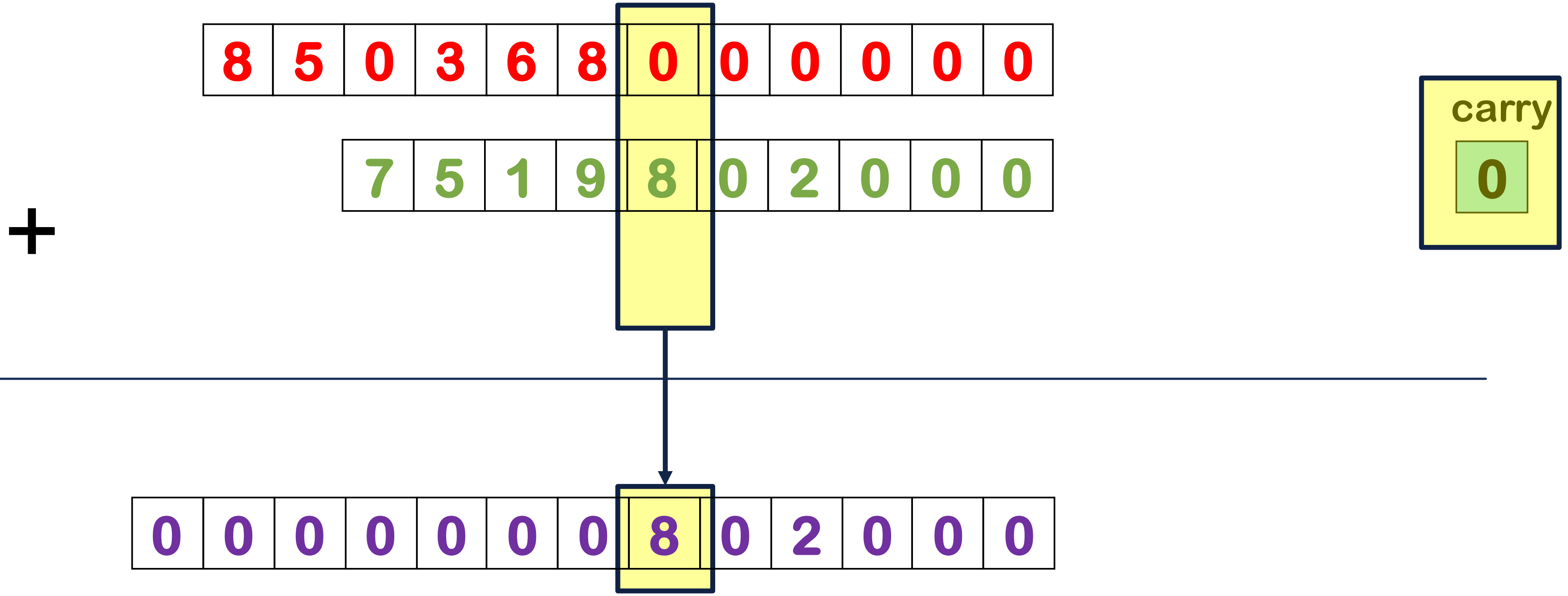
7	5	1	9	8	0	2	0	0	0

carry

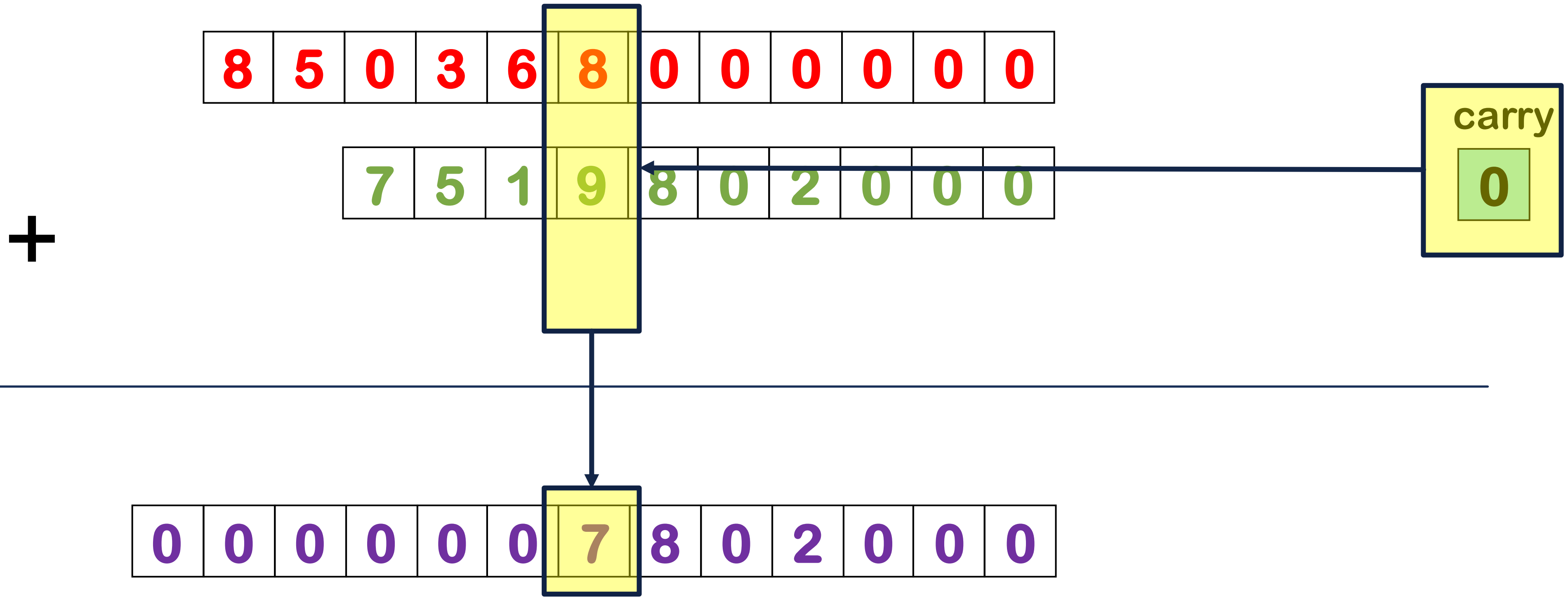
0

0	0	0	0	0	0	0	0	0	2	0	0	0

L.I.M: SUMMATION



L.I.M: SUMMATION



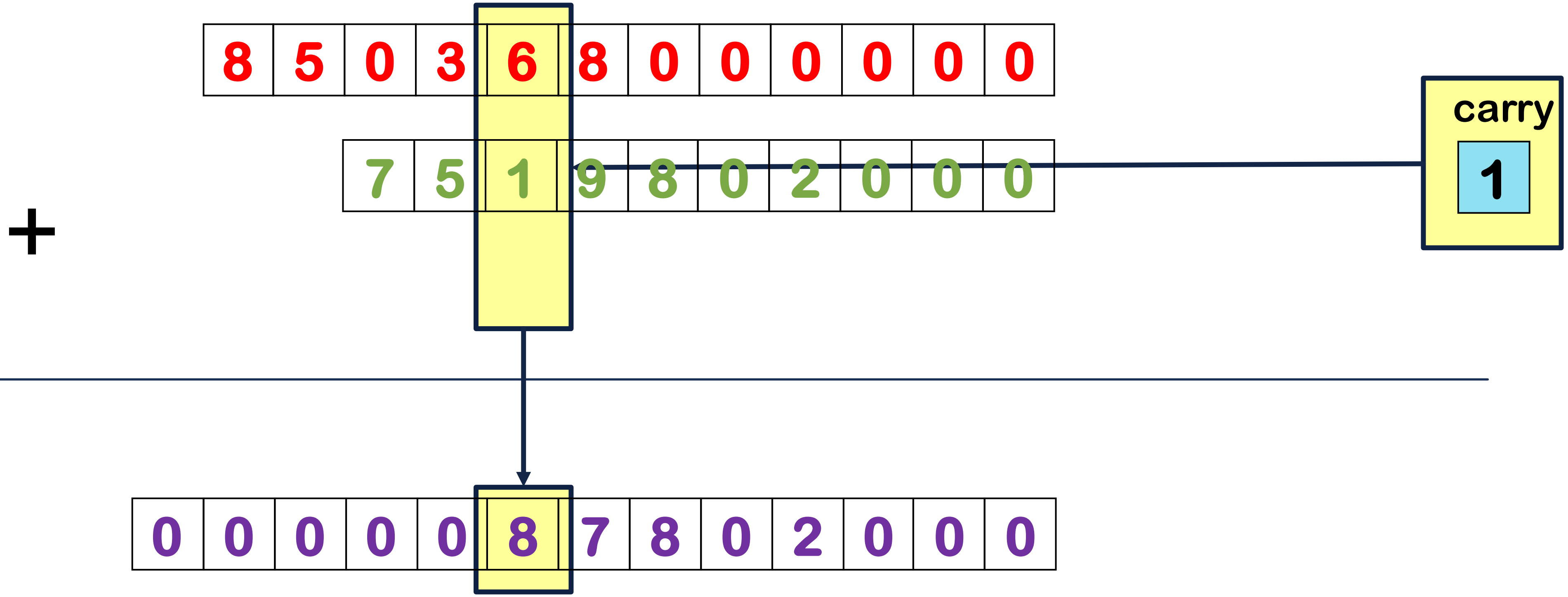
L.I.M: SUMMATION

	8	5	0	3	6	8	0	0	0	0	0	0
+			7	5	1	9	8	0	2	0	0	0
0	0	0	0	0	0	7	8	0	2	0	0	0

carry

1

L.I.M: SUMMATION



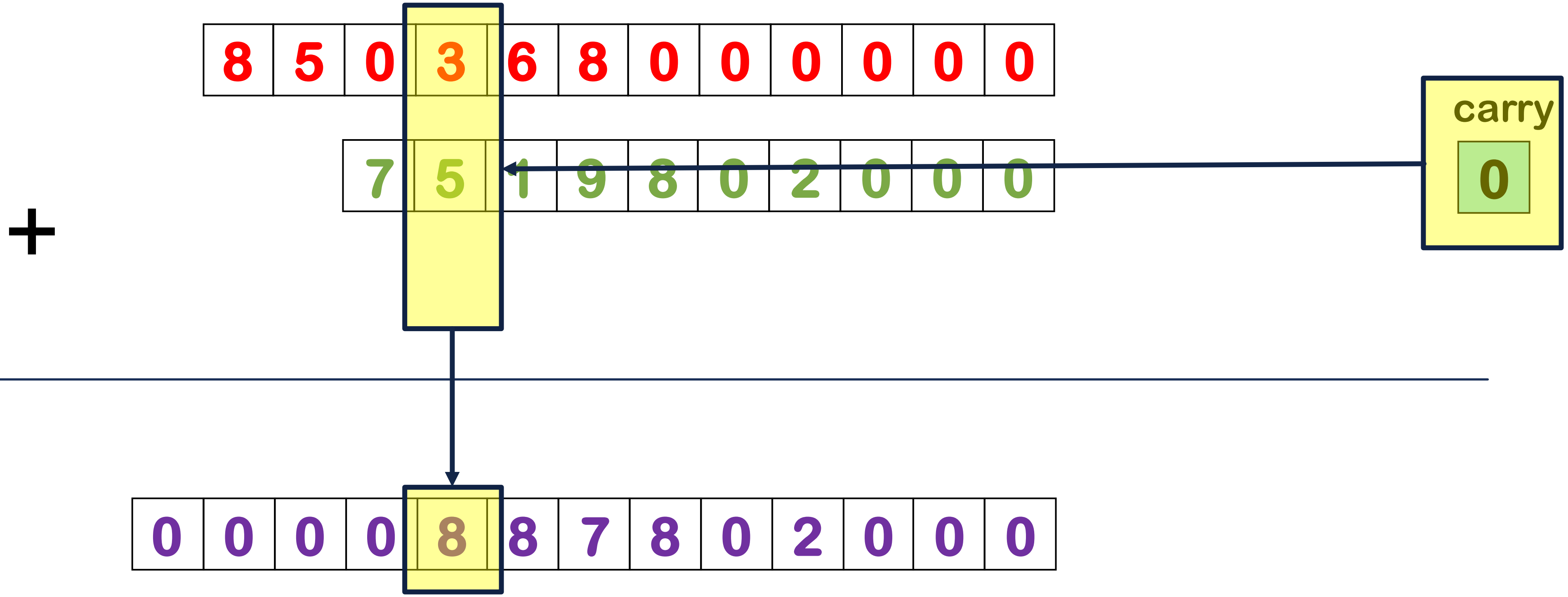
L.I.M: SUMMATION

	8	5	0	3	6	8	0	0	0	0	0	0
+			7	5	1	9	8	0	2	0	0	0
0	0	0	0	0	8	7	8	0	2	0	0	0

carry

0

L.I.M: SUMMATION



L.I.M: SUMMATION

	8	5	0	3	6	8	0	0	0	0	0	0
+			7	5	1	9	8	0	2	0	0	0
0	8	5	7	8	8	7	8	0	2	0	0	0

carry

0

$O(n)$

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$8,256,734 * 103,875$$

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$8,256,734 * 103,875 = (8,256 * 1000 + 734) * (103 * 1000 + 875)$$

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$8,256,734 * 103,875 = (\boxed{8256 * 1000} + 734) * (\boxed{103 * 1000} + 875)$$

$$= \boxed{(8256 * 103) * 10^6} +$$

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$8,256,734 * 103,875 = (\boxed{8256 * 1000} + 734) * (103 * 1000 + \boxed{875})$$

$$= (8256 * 103) * 10^6 + \boxed{(8256 * 875) * 10^3} +$$

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$8,256,734 * 103,875 = (8256 * 1000 + \boxed{734}) * (\boxed{103 * 1000} + 875)$$

$$= (8256 * 103) * 10^6 + (8256 * 875) * 10^3 + \boxed{(734 * 103) * 10^3} +$$

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$8,256,734 * 103,875 = (8256 * 1000 + \boxed{734}) * (103 * 1000 + \boxed{875})$$

$$= (8256 * 103) * 10^6 + (8256 * 875) * 10^3 + (734 * 103) * 10^3 + \boxed{(734 * 875)}$$

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$8,256,734 * 103,875 = (8256 * 1000 + 734) * (103 * 1000 + 875)$$

$$= (8256 * 103) * 10^6 + \boxed{(8256 * 875) * 10^3 + (734 * 103) * 10^3} + (734 * 875)$$

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$\mathbf{8,256,734 * 103,875 = (8256 * 1000 + 734) * (103 * 1000 + 875)}$$

$$\mathbf{= (8256 * 103) * 10^6 + \boxed{(8256 * 875 + 734 * 103) * 10^3} + (734 * 875)}$$

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$8,256,734 * 103,875 = (8256 * 1000 + 734) * (103 * 1000 + 875)$$

Not multiplication

$$= (8256 * 103) * 10^6 + (8256 * 875 + 734 * 103) * 10^3 + (734 * 875)$$

$\boxed{4}$

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

- u and v are two n-digit large integer:

$$\boldsymbol{u = x * 10^m + y} \quad , \qquad \boldsymbol{v = w * 10^m + z} \qquad \boldsymbol{m = \left\lfloor \frac{n}{2} \right\rfloor}$$

$$u * v = (x * w) * 10^{2m} + (x * z + y * w) * 10^m + y * z$$

Example:

$$7{,}824{,}657 \times 64{,}528 \rightarrow 7824657 \times 0064528 \qquad m = 4$$

L.I.M: ALGORITHM

```
multiply(u,v)
    if(u==‘0’ or v==‘0’) return ‘0’
    n1=u.size()
    n2=v.size()
    n=max(n1,n2)
    if(n<threshold) return str(int(u)*int(v))
    m=n/2
    x,y=split(u)
    w,z=split(v)
    p=multiply(x,w)
    for(i=0;i<2*m;i++) p=p+’0’
    t1=multiply(x,z)
    t2=multiply(y,w)
    s=sum(t1,t2)
    for(i=0;i<m;i++) s=s+’0’
    q=multiply(y,z)
    return sum(sum(p,s),q)
```

L.I.M: ALGORITHM

```
multiply(u,v)
   if(u==‘0’ or v==‘0’) return ‘0’
   n1=u.size()
   n2=v.size()
   n=max(n1,n2)
   if(n<threshold) return str(int(u)*int(v))
   m=n/2
   x,y=split(u)
   w,z=split(v)
   p=multiply(x,w)
   for(i=0;i<2*m;i++) p=p+’0’
   t1=multiply(x,z)
   t2=multiply(y,w)
   s=sum(t1,t2)
   for(i=0;i<m;i++) s=s+’0’
   q=multiply(y,z)
   return sum(sum(p,s),q)
```

L.I.M: ALGORITHM

```
multiply(u,v)
   if(u=='0' or v=='0') return '0'
   n1=u.size()
   n2=v.size()
   n=max(n1,n2)
   if(n<threshold) return str(int(u)*int(v))
   m=n/2
   x,y=split(u)
   w,z=split(v)
   p=multiply(x,w)
   for(i=0;i<2*m;i++) p=p+'0'
   t1=multiply(x,z)
   t2=multiply(y,w)
   s=sum(t1,t2)
   for(i=0;i<m;i++) s=s+'0'
   q=multiply(y,z)
   return sum(sum(p,s),q)
```

L.I.M: ALGORITHM

```
multiply(u,v)
    if(u=='0' or v=='0') return '0'
    n1=u.size()
    n2=v.size()
    n=max(n1,n2)
    if(n<threshold) return str(int(u)*int(v))
    m=n/2
    x,y=split(u)
    w,z=split(v)
    p=multiply(x,w)
    for(i=0;i<2*m;i++) p=p+'0'
    t1=multiply(x,z)
    t2=multiply(y,w)
    s=sum(t1,t2)
    for(i=0;i<m;i++) s=s+'0'
    q=multiply(y,z)
    return sum(sum(p,s),q)
```

L.I.M: ALGORITHM

```
multiply(u,v)
    if(u=='0' or v=='0') return '0'
    n1=u.size()
    n2=v.size()
    n=max(n1,n2)
    if(n<threshold) return str(int(u)*int(v))
    m=n/2
    x,y=split(u)
    w,z=split(v)
    p=multiply(x,w)
    for(i=0;i<2*m;i++) p=p+'0'
    t1=multiply(x,z)
    t2=multiply(y,w)
    s=sum(t1,t2)
    for(i=0;i<m;i++) s=s+'0'
    q=multiply(y,z)
    return sum(sum(p,s),q)
```

L.I.M: ALGORITHM

```
multiply(u,v)
    if(u=='0' or v=='0') return '0'
    n1=u.size()
    n2=v.size()
    n=max(n1,n2)
    if(n<threshold) return str(int(u)*int(v))
    m=n/2
    x,y=split(u)
    w,z=split(v)
    p=multiply(x,w)
    for(i=0;i<2*m;i++) p=p+'0'
    t1=multiply(x,z)
    t2=multiply(y,w)
    s=sum(t1,t2)
    for(i=0;i<m;i++) s=s+'0'
    q=multiply(y,z)
    return sum(sum(p,s),q)
```

L.I.M: ALGORITHM

```
multiply(u,v)
    if(u=='0' or v=='0') return '0'
    n1=u.size()
    n2=v.size()
    n=max(n1,n2)
    if(n<threshold) return str(int(u)*int(v))
    m=n/2
    x,y=split(u)
    w,z=split(v)
    p=multiply(x,w)
    for(i=0;i<2*m;i++) p=p+'0'
    t1=multiply(x,z)
    t2=multiply(y,w)
    s=sum(t1,t2)
    for(i=0;i<m;i++) s=s+'0'
    q=multiply(y,z)
    return sum(sum(p,s),q)
```

L.I.M: ALGORITHM

```
multiply(u,v)
    if(u=='0' or v=='0') return '0'
    n1=u.size()
    n2=v.size()
    n=max(n1,n2)
    if(n<threshold) return str(int(u)*int(v))
    m=n/2
    x,y=split(u)
    w,z=split(v)
    p=multiply(x,w)
    for(i=0;i<2*m;i++) p=p+'0'
    t1=multiply(x,z)
    t2=multiply(y,w)
    s=sum(t1,t2)
    for(i=0;i<m;i++) s=s+'0'
    q=multiply(y,z)
    return sum(sum(p,s),q)
```

$$T(n) = O(n) + 4T\left(\frac{n}{2}\right)$$

`multiply(array u,v)`	$T(n)$
`if(u=='0' or v=='0') return '0'`	$O(1)$
`n1=u.size()`	
`n2=v.size()`	$O(1)$
`n=max(n1,n2)`	
`if(n<threshold) return str(int(u)*int(v))`	$T(1) = 1$
`m=n/2`	
`x,y=split(u)`	$O(1)$
`w,z=split(v)`	
`p=multiply(x,w)`	$T\left(\frac{n}{2}\right)$
`for(i=0;i<2*m;i++) p=p+'0'`	$O(n)$
`t1=multiply(x,z)`	$T\left(\frac{n}{2}\right)$
`t2=multiply(y,w)`	$T\left(\frac{n}{2}\right)$
`s=sum(t1,t2)`	$O(n)$
`for(i=0;i<m;i++) s=s+'0'`	$O(n)$
`q=multiply(y,z)`	$T\left(\frac{n}{2}\right)$
`return sum(sum(p,s),q`	$O(n)$

TIME COMPLEXITY-MASTER THEOREM

$$T(n) = O(n) + 4T\left(\frac{n}{2}\right)$$

$$a = 4, b = 2, k = 1, p = 0$$

$$T(n) = O\left(n^{\log_2 4}\right) = O(n^2)$$

$$T(n) = a\,T\left(\frac{n}{b}\right) + \theta(n^k \log^p n)$$

$$if\ a > b^k, then\ T(n) = \theta\left(n^{\log_b a}\right)$$

LIM: MODIFICATION

$$u * v = (x * w) * 10^{2m} + \left(\boxed{x * z + y * w}\right) * 10^{m} + y * z$$

LIM: MODIFICATION

$$u * v = (x * w) * 10^{2m} + (x * z + y * w) * 10^{m} + y * z$$

$$x * z + y * w = (x + y) * (z + w) - (x * w) - (y * z)$$

$$(\boldsymbol{x} + \boldsymbol{y})(\boldsymbol{z} + \boldsymbol{w}) =$$

$$\boldsymbol{xz} + \boldsymbol{xw} + \boldsymbol{yz} + \boldsymbol{yw} =$$

$$(\boldsymbol{xz} + \boldsymbol{yw}) + (\boldsymbol{xw}) + (\boldsymbol{yz})$$

LIM: MODIFICATION

$$\cancel{u * v = (x * w) * 10^{2m} + (x * z + y * w) * 10^{m} + y * z}$$

$$x * z + y * w = (x + y) * (z + w) - (x * w) - (y * z)$$

$$r = (x + y) * (z + w), \quad p = x * w, \quad q = y * z$$

$$u * v = p * 10^{2m} + (r - p - q) * 10^{m} + q$$

L.I.M: ALGORITHM

```
multiply(array u,v)
    if(u=='0' or v=='0') return '0'
    n1=u.size()
    n2=v.size()
    n=max(n1,n2)
    if(n<threshold) return str(int(u)*int(v))
    m=n/2
    x,y=split(u)
    w,z=split(v)
    p=multiply(x,w)
    for(i=0;i<2*m;i++) p=p+'0'
    q=multiply(y,z)
    s1=sum(x,y)
    s2=sum(w,z)
    r=multiply(s1,s2)
    s=sum(r,-sum(p,q))
    for(i=0;i<m;i++) s=s+'0'
    return sum(sum(p,s),q)
```

```
multiply(array u,v)
    if(u=='0' or v=='0') return '0'
    n1=u.size()
    n2=v.size()
    n=max(n1,n2)
    if(n<threshold) return str(int(u)*int(v))
    m=n/2
    x,y=split(u)
    w,z=split(v)
    p=multiply(x,w)
    for(i=0;i<2*m;i++) p=p+'0'
    q=multiply(y,z)
    s1=sum(x,y)
    s2=sum(w,z)
    r=multiply(s1,s2)
    s=sum(r,-sum(p,q))
    for(i=0;i<m;i++) s=s+'0'
    return sum(sum(p,s),q
```

$T(n)$

$O(1)$

$O(1)$

$T(1)=1$

$O(1)$

$T\left(\frac{n}{2}\right)$

$O(n)$

$T\left(\frac{n}{2}\right)$

$O(n)$

$T\left(\frac{n}{2}\right)$

$O(n)$

$O(n)$

$O(n)$

$$\boxed{T(n) = O(n) + 3T\left(\frac{n}{2}\right)}$$

TIME COMPLEXITY-MASTER THEOREM

$$T(n) = O(n) + 3T\left(\frac{n}{2}\right)$$

$a = 3, b = 2, k = 1, p = 0$

$T(n) = O\left(n^{\log_2 3}\right) = O(n^{1.58})$

$$T(n) = a\,T\left(\frac{n}{b}\right) + \theta(n^k \log^p n)$$

$if\ a > b^k, then\ T(n) = \theta\left(n^{\log_b a}\right)$

ANY QUESTION??

HAVE A NICE DAY!